Survey Narrative:

Reporting New Trends in Software Development Methodologies

CLO & PLO Alignment

- ***CLO-3****: Report new trends in software development methodologies*
- ***Mapped PLO-12****: Lifelong Learning – Ability to recognize the importance of and pursue lifelong learning in the broader context of innovation and technological developments*
- ***Learning Level****: A3*

1. Introduction to the Activity

This activity evaluates students' ability to identify, analyze, and report emerging software development practices. By interacting with industry professionals, students will learn how real-world teams adapt methodologies, standards, and tools to meet evolving challenges.

It ensures the attainment of CLO-3 → PLO-12 by developing skills for lifelong learning and adaptation.

2. Instructions

1. Work in existing groups.
2. Visit or virtually connect with a software house/IT company.
3. Conduct interviews using the Survey Questionnaire.
4. Analyze the collected data to extract key trends, comparisons, and insights.
5. Prepare a written Analytical Report. Also attach the Raw data of your survey, along with the report

3. Survey Questionnaire (CLO-3 → PLO-12)

The questionnaire includes sections on methodologies, architecture, coding, testing, and maintenance.

Section 1 – Respondent Profile

- Organization: _____
- Respondent Role: Developer / Tester / Architect / Manager
- Years of Experience: _____
- Tools Used (Jira, GitHub, CI/CD, etc.): _____

Section 2 – Development Methodologies
1. Which software development lifecycle model(s) do you commonly use?
2. What factors drive your choice of model (e.g., project type, client requirements, deadlines)?
3. Do you adopt Agile or hybrid approaches in large/critical projects? Why?
4. How do you adapt methodologies when handling safety-critical systems vs. enterprise systems?

Section 3 – Architecture & Design Practices
1. Is architecture formally created before implementation?
2. Which architectural styles or patterns are trending in your organization?
3. Are design patterns regularly used in practice? If yes, which ones?

Section 4 – Coding & Standards
1. Are coding standards enforced? At what level (industry/organization/project)?
2. Which code review practices are in place?
3. How do you ensure quality, cohesion, and maintainability in code?

Section 5 – Testing & Quality Assurance
1. Which types of testing (unit, integration, UAT, automated testing) are trending in your projects?
2. What tools and CI/CD pipelines are used for quality assurance?
3. Is test suite optimization or automated regression testing widely practiced?

Section 6 – Maintenance & Sustainability
1. What are the most common challenges faced during software maintenance?
2. How do you incorporate future-proofing and sustainability into projects?
3. What emerging practices (e.g., DevOps, AI-assisted testing, cloud-native pipelines) do you foresee dominating the future?

4. Analytical Report Outline

The report must be structured, evidence-based, and analytical. Students should follow this outline:

1. **Title Page** – Course Code (EC-360), Group Members, Date, Survey Title
2. **Executive Summary** – Objectives, methodology, and key findings
3. **Introduction** – Purpose of the survey, CLO-3 and PLO-12 linkage
4. **Survey Methodology** – Organization(s) visited, professionals interviewed, tools used
5. **Findings & Analysis** – Methodologies, architecture, coding, testing, maintenance
6. **Discussion of Trends** – Emerging practices, comparison with textbook methods, lifelong learning link
7. **Conclusion** – Key insights and student reflection on lifelong learning

8. **References** – Books, research papers, industry blogs
9. **Appendices** – Raw survey responses (mandatory)

5. Rubric for Survey & Analytical Report

Criterion	Excellent (A)	Good (B)	Satisfactory (C)	Poor (D/F)
Survey Execution (20%)	Multiple professionals interviewed, wide coverage, detailed responses	Adequate responses, most questions covered	Limited professionals or incomplete data	Minimal effort, poor data collection
Report Structure & Clarity (15%)	Follows structured outline, well-organized, professional	Structured but minor gaps	Some structure but lacks flow	Unstructured, unclear
Findings & Analysis (35%)	Critical insights, clear trends identified, strong comparisons	Good insights, moderate comparisons	Basic reporting, weak analytical depth	Only descriptive, no critical analysis
Discussion of Trends (15%)	Excellent linkage with emerging practices & lifelong learning	Some linkage, moderate depth	Limited linkage	No linkage, purely descriptive
Conclusion & Reflection (10%)	Strong reflection on lifelong learning and CLO-3 attainment	Moderate reflection, partial linkage	Weak reflection, generic	No reflection
References & Evidence (5%)	Properly cited, professional formatting	Adequate references, minor issues	Few references, weak support	No references provided